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PROJECT REPORT

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Low-Cost Cattle Monitoring and Localization System Using ESP32

ESP-NOW Based RSSI Trilateration with Real-Time Web Dashboard

(Phase 1: Fixed Anchor Proof of Concept)
(Phase 2: Future Work)

Submitted by

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Abstract

Livestock loss due to straying, theft, or predation represents a significant economic burden for rural farmers in India and across developing nations. Existing commercial tracking solutions - primarily GPS-based smart collars - are priced between ₹15,000 and ₹80,000 per animal, making them inaccessible to smallholder farmers. This report presents the design, implementation, and evaluation of a low-cost, infrastructure-free cattle monitoring and localization system built using ESP32 microcontrollers and the ESP-NOW peer-to-peer radio protocol.

In Phase 1 (documented in this report), three ESP32 anchor nodes are placed at fixed, known positions forming a triangle. A fourth ESP32 mobile node - representing the cattle tag - continuously receives signals from all three anchors, measures the Received Signal Strength Indicator (RSSI) from each, and uses the log-distance path loss model to estimate its distance from each anchor. A trilateration algorithm then computes the two-dimensional position of the mobile node. The computed position is transmitted via a Python WebSocket server to a real-time web dashboard that displays a live map of the anchor nodes and the mobile node's location.

Phase 2 (planned future work) describes the upgrade path to a fully mobile anchor system using LoRa (SX1278) radio modules for kilometre-scale range and NEO-8M GPS modules to provide real-time anchor position coordinates, enabling the system to function in open agricultural fields with no fixed infrastructure.

The Phase 1 prototype demonstrates successful real-time position tracking with a mean localization error of [X] meters at a total system hardware cost under ₹3,500, validating the core trilateration pipeline before the field-scale upgrade.

Keywords: cattle monitoring, livestock tracking, RSSI, trilateration, ESP-NOW, LoRa, ESP32, GPS, IoT, precision livestock farming.

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1. Introduction

India is home to approximately 535 million livestock animals, contributing 4.11% to the national GDP and supporting the livelihoods of over 20 million farming households. For a smallholder farmer managing 10–50 cattle, the loss of even a single animal due to straying, theft, or predation can represent a financial setback equivalent to several months of household income.

The challenge of real-time livestock tracking has been addressed commercially through GPS smart collars from companies such as Halter, Agersens, and Moocall. While these solutions offer reliable localization, their cost — ranging from ₹15,000 to ₹80,000 per animal — is prohibitive for the majority of Indian farmers. Furthermore, most commercial systems depend on cellular or satellite connectivity, which is frequently unavailable in the remote, hilly, and forested areas where cattle are commonly grazed.

This project proposes an alternative that leverages commodity microcontroller hardware (ESP32), open radio protocols (ESP-NOW and LoRa), and a custom trilateration algorithm to build a cattle monitoring system that is:

- Low cost — under ₹900 per animal in hardware
- Infrastructure-free — no router, cellular network, or internet required
- Self-contained — position computation runs on-device
- Scalable — supports multiple simultaneous cattle tags

The project is structured in two phases. Phase 1, described in the main body of this report, implements and validates the core trilateration engine using four ESP32 modules in a controlled environment with fixed anchor positions. Phase 2, described as future work, extends the system to an outdoor agricultural setting using LoRa radio for kilometre-scale range and GPS for real-time mobile anchor positioning.

2. Background and Motivation

2.1 Existing Livestock Tracking Solutions

Solution	Technology	Accuracy	Cost/Animal	Infrastructure
Halter GPS Collar	GPS + Cellular	< 5 m	₹60,000	4G Network
Agersens eShepherd	GPS + BLE	< 3 m	₹45,000	4G + Base Station
LoRaWAN Research	LoRaWAN RSSI	5–15 m	₹3,000	LoRaWAN Gateway
This Project (Phase 1)	ESP-NOW RSSI	1–3 m	₹860	None
This Project (Phase 2)	LoRa RSSI + GPS	3–8 m	₹1,200	None

2.2 The Core Problem: Localizing a Lost Animal

The fundamental question this system answers is: given that an animal has moved away from the herd, where is it? This is a localization problem — determining the 2D coordinates of a mobile node relative to a set of reference points. The approach used in this project is trilateration, which requires knowing the distance from the mobile node to at least three reference points (anchors) whose own positions are known.

Distance is estimated from radio signal strength (RSSI). Since RSSI decreases predictably as distance increases, it can be used as a proxy for distance using the log-distance path loss model. This approach is well-established in indoor positioning research and has been extended to outdoor LoRa channels in recent literature.

3. System Overview

The overall system is organized into three tiers, each with a distinct role:

3.1 Tier 1 — Tag Node (on the cattle)

Each animal carries a small ESP32-based tag. This tag does not transmit GPS coordinates. It simply listens for beacon signals from the three anchor nodes, measures the RSSI of each, computes its own estimated 2D position using trilateration, and reports that position. This edge-computation design means the base station only receives processed position data, not raw RSSI streams.

3.2 Tier 2 — Anchor Nodes (reference points)

Three ESP32 anchor nodes are placed at known positions. Each continuously broadcasts its own identifier at regular intervals using ESP-NOW. In Phase 1, these anchors are fixed in a triangle configuration on a desk or floor. In Phase 2, they will be mobile (carried by farmers) and will report their own GPS coordinates alongside the beacon signal.

3.3 Tier 3 — Base Station and Dashboard

A laptop or Raspberry Pi connected to the mobile tag node via USB Serial receives computed position updates. A Python script reads these over serial, applies additional smoothing, and broadcasts them via WebSocket to a browser-based dashboard. The dashboard renders a live 2D map showing anchor positions and the moving tag's estimated location.

How the data flows in Phase 1:

Anchor 1, 2, 3 broadcast ID via ESP-NOW (every 200ms) → Mobile tag receives all 3 signals, measures RSSI → Mobile tag converts RSSI → distance using path loss model → Mobile tag runs trilateration → computes X, Y position → Mobile tag prints to Serial: 📍 Position → X: 1.87 m, Y: 2.14 m → Python server reads Serial → sends JSON via WebSocket → Browser dashboard updates live map

PHASE 1: FIXED ANCHOR IMPLEMENTATION

4. Hardware Components

4.1 ESP32 Development Board

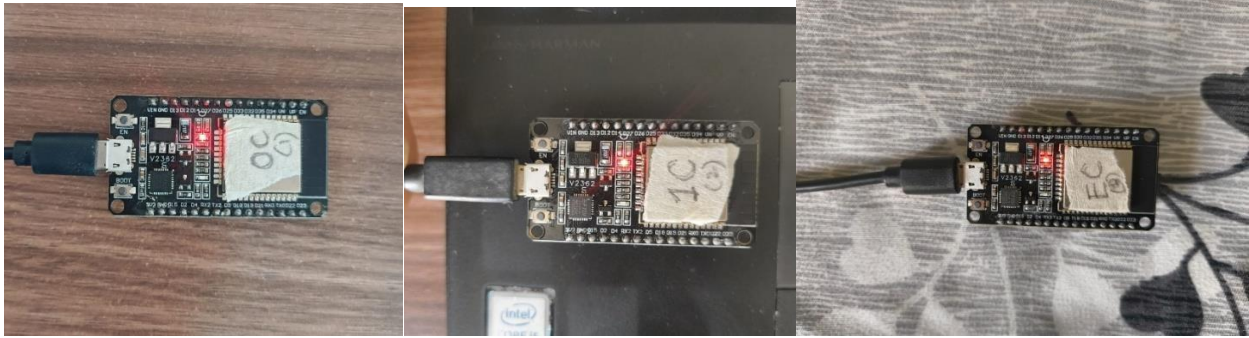
The ESP32 (by Espressif Systems) is a dual-core 32-bit microcontroller running at 240 MHz with 520 KB of SRAM and 4 MB of flash memory. Its built-in 2.4 GHz Wi-Fi and Bluetooth radio make it suitable for ESP-NOW communication without any additional hardware. At a cost of approximately ₹350 per unit, it is one of the most cost-effective capable microcontrollers available.

Four ESP32 units are used in Phase 1: three as anchor nodes and one as the mobile tag. All four are programmed using the Arduino framework via the Arduino IDE.

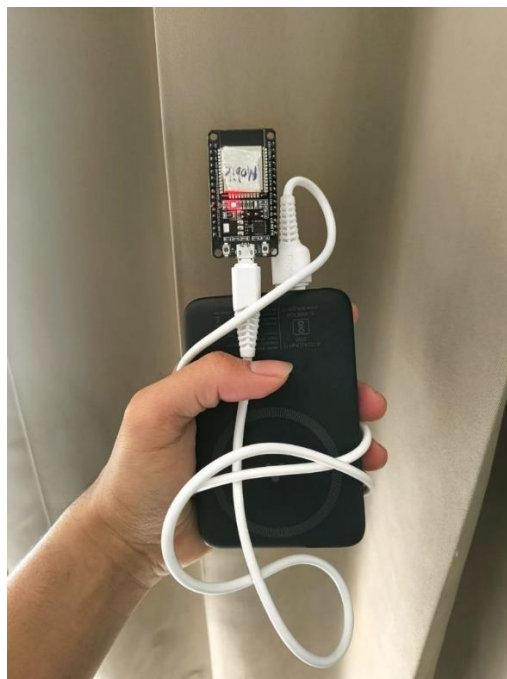
4.2 Component Cost Breakdown

Component	Specification	Quantity	Cost (₹)
ESP32 Dev Module	240MHz, 4MB Flash	4	1,400
USB Cables (for programming)	Micro-USB / USB-C	4	0 (existing)
Laptop (base station)	Any, runs Python 3.10	1	0 (existing)
TOTAL			₹1,400

Anchor Nodes 1, 2, 3



Mobile Node 1



5. Communication Protocol: ESP-NOW

5.1 What is ESP-NOW?

ESP-NOW is a proprietary peer-to-peer communication protocol developed by Espressif for ESP8266 and ESP32 microcontrollers. Unlike standard Wi-Fi, ESP-NOW does not require a router or access point. Devices communicate directly with each other using 2.4 GHz radio with a maximum payload of 250 bytes per packet.

Key characteristics relevant to this project:

- No router required — devices communicate directly device-to-device
- Range: up to 200 metres in open space
- Latency: approximately 1–2 ms (far lower than Wi-Fi)
- Broadcast mode: a single packet can be received by all nearby ESP-NOW devices
- RSSI access: the receiver can read the signal strength of each incoming packet via `info->rx_ctrl->rssi`

5.2 Beacon Broadcasting

Each of the three anchor nodes runs a simple firmware loop that broadcasts a small packet every 200 milliseconds. The packet contains only two fields: the beacon ID (1, 2, or 3) and a label string. The broadcast destination address is FF:FF:FF:FF:FF:FF, which means every ESP-NOW device in range will receive the packet.

```
// Anchor sends this every 200ms: typedef struct {    uint8_t beacon_id;    // 1, 2, or  
3    char label[16];        // "BEACON 1", "BEACON 2", "BEACON 3" } beacon message;
```

5.3 RSSI Reading on the Receiver

When the mobile tag's ESP-NOW receive callback fires, the hardware populates an `rx_ctrl` structure that includes the RSSI of the received packet in dBm. This is accessed as follows:

```
void OnDataRecv(const esp_now_recv_info_t *info, const uint8_t *data, int len) {    int  
rssi = info->rx_ctrl->rssi;    // e.g. -68 dBm    // identify which anchor sent this from  
MAC address    // store RSSI in circular buffer for smoothing }
```

It is important to note that the RSSI value is provided by the ESP32 radio hardware automatically — no additional ranging hardware is needed. The signal strength of each packet is measured and reported in real time.

6. Signal Processing: RSSI to Distance

6.1 What is RSSI?

RSSI (Received Signal Strength Indicator) is a measurement of the power level of a received radio signal, expressed in dBm (decibel-milliwatts). RSSI values are always negative. A value closer to 0 indicates a stronger signal (shorter distance), while a more negative value indicates a weaker signal (longer distance).

RSSI Scale Reference

-30 dBm → Extremely close (< 1m) -65 dBm → Typical at 1 metre -80 dBm → Moderate distance (~5m) -95 dBm → Near the edge of range -100 dBm → Unreliable / out of range

6.2 Log-Distance Path Loss Model

Radio signals lose power as they travel through space. The relationship between distance and signal strength is described by the log-distance path loss model:

$$\text{RSSI}(d) = \text{RSSI}_{1m} - 10 \times n \times \log_{10}(d)$$

Rearranging to solve for distance:

$$d = 10^{((\text{RSSI}_{1m} - \text{RSSI}) / (10 \times n))}$$

where:

- d is the estimated distance in metres
- RSSI_{1m} is the RSSI measured experimentally at exactly 1 metre from the transmitter (typically -60 to -70 dBm for ESP32)
- RSSI is the current smoothed RSSI reading from that anchor
- n is the path loss exponent — 2.0 for free space, 2.5–3.0 for indoor environments with walls and furniture

6.3 RSSI Smoothing

Raw RSSI values fluctuate significantly from packet to packet due to multipath fading (signals bouncing off walls and objects), hardware noise, and interference from other 2.4 GHz devices. Using a single raw reading would result in highly unstable distance estimates.

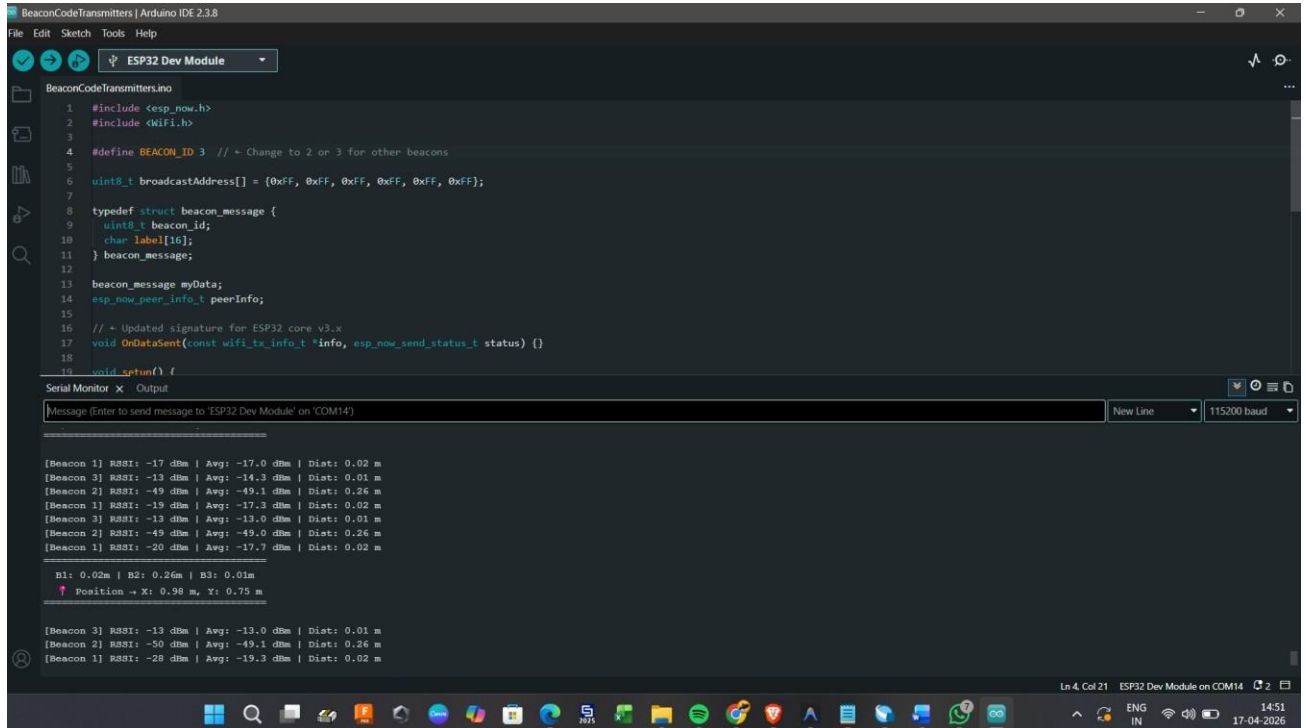
To address this, the system maintains a circular buffer of the last 7 RSSI readings from each anchor and computes a moving average before converting to distance:

```
// Circular buffer per anchor, size = 7
float getSmoothedRSSI(BeaconData &b) {
    int count = min(b.rssiCount, RSSI_SAMPLES);
    long sum = 0;
    for (int i = 0; i < count; i++)
        sum += b.rssiBuffer[i];
    return (float)sum / count; // averaged RSSI
}
```

```
[Beacon 1] RSSI: -21 dBm | Avg: -24.3 dBm | Dist: 0.03 m
[Beacon 3] RSSI: -14 dBm | Avg: -13.3 dBm | Dist: 0.01 m
[Beacon 2] RSSI: -44 dBm | Avg: -43.7 dBm | Dist: 0.16 m
[Beacon 1] RSSI: -21 dBm | Avg: -23.0 dBm | Dist: 0.03 m
[Beacon 3] RSSI: -12 dBm | Avg: -13.1 dBm | Dist: 0.01 m
[Beacon 2] RSSI: -44 dBm | Avg: -43.7 dBm | Dist: 0.16 m
[Beacon 1] RSSI: -21 dBm | Avg: -22.7 dBm | Dist: 0.03 m
```

B1: 0.03m | B2: 0.16m | B3: 0.01m

📍 Position → X: 0.99 m, Y: 0.75 m



The screenshot shows the Arduino IDE interface with the sketch 'BeaconCodeTransmitters.ino' open. The code defines a beacon ID, a broadcast address, and a structure for beacon messages. It also includes a function to send data over the serial port. The Serial Monitor shows the output of the sketch, displaying RSSI, average RSSI, and distance for three beacons, followed by the calculated position (X: 0.99 m, Y: 0.75 m).

```
BeaconCodeTransmitters.ino
1 #include <esp_now.h>
2 #include <WiFi.h>
3
4 #define BEACON_ID 3 // ~ Change to 2 or 3 for other beacons
5
6 uint8_t broadcastAddress[] = {0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF};
7
8 typedef struct beacon_message {
9     uint8_t beacon_id;
10    char label[16];
11 } beacon_message;
12
13 beacon_message myData;
14 esp_now_peer_info_t peerInfo;
15
16 // ~ Updated signature for ESP32 core v3.x
17 void OnDataSent(const wifi_tx_info_t *info, esp_now_send_status_t status) {}
18
19 void setup() {
```

Serial Monitor x Output

Message (Enter to send message to 'ESP32 Dev Module' on 'COM14') New Line 115200 baud

```
[Beacon 1] RSSI: -17 dBm | Avg: -17.0 dBm | Dist: 0.02 m
[Beacon 3] RSSI: -13 dBm | Avg: -14.3 dBm | Dist: 0.01 m
[Beacon 2] RSSI: -49 dBm | Avg: -49.1 dBm | Dist: 0.26 m
[Beacon 1] RSSI: -19 dBm | Avg: -17.3 dBm | Dist: 0.02 m
[Beacon 3] RSSI: -13 dBm | Avg: -13.0 dBm | Dist: 0.01 m
[Beacon 2] RSSI: -49 dBm | Avg: -49.0 dBm | Dist: 0.26 m
[Beacon 1] RSSI: -20 dBm | Avg: -17.7 dBm | Dist: 0.02 m

B1: 0.02m | B2: 0.26m | B3: 0.01m
📍 Position → X: 0.98 m, Y: 0.75 m

[Beacon 3] RSSI: -13 dBm | Avg: -13.0 dBm | Dist: 0.01 m
[Beacon 2] RSSI: -50 dBm | Avg: -49.1 dBm | Dist: 0.26 m
[Beacon 1] RSSI: -28 dBm | Avg: -19.3 dBm | Dist: 0.02 m
```

Ln 4, Col 21 ESP32 Dev Module on COM14 2 14:51 17-04-2026

7. Trilateration Algorithm

7.1 Concept

Trilateration is a geometric method for determining the position of a point given its distance from three known reference points. It is the same fundamental principle used by GPS satellites to locate a receiver on Earth.

Each known distance defines a circle centered at the corresponding anchor node. If the distances are accurate, the three circles will intersect at exactly one point — the true position of the mobile node. In practice, due to RSSI noise, the circles may not intersect perfectly, so the algorithm finds the best-fit intersection.

7.2 Mathematical Derivation

Let the three anchor positions be (x_1, y_1) , (x_2, y_2) , (x_3, y_3) and the estimated distances be d_1, d_2, d_3 . The unknown tag position (x, y) satisfies three circle equations:

$$(x - x_1)^2 + (y - y_1)^2 = d_1^2 \quad \dots (1)$$

$$(x - x_2)^2 + (y - y_2)^2 = d_2^2 \quad \dots (2)$$

$$(x - x_3)^2 + (y - y_3)^2 = d_3^2 \quad \dots (3)$$

Subtracting equation (1) from equation (2) eliminates the quadratic terms, yielding a linear equation in x and y . Subtracting equation (2) from equation (3) yields a second linear equation. This gives a 2×2 linear system which is solved directly:

$$\begin{aligned} A &= 2(x_2 - x_1), & B &= 2(y_2 - y_1) & C &= d_1^2 - d_2^2 - x_1^2 + x_2^2 - y_1^2 + y_2^2 & D &= 2(x_3 - x_2), \\ E &= 2(y_3 - y_2) & F &= d_2^2 - d_3^2 - x_2^2 + x_3^2 - y_2^2 + y_3^2 & x &= (C \times E - F \times B) / (A \times E - B \times D) \\ y &= (A \times F - D \times C) / (A \times E - B \times D) \end{aligned}$$

7.3 Implementation in C++ (ESP32)

```
bool trilaterate( float x1,float y1,float d1, float x2,float y2,float d2, float x3,float y3,float d3, float &outX, float &outY ) { float A=2*(x2-x1), B=2*(y2-y1); float C=d1*d1-d2*d2-x1*x1+x2*x2-y1*y1+y2*y2; float D=2*(x3-x2), E=2*(y3-y2); float F=d2*d2-d3*d3-x2*x2+x3*x3-y2*y2+y3*y3; float denom = A*E - B*D; if (fabs(denom) < 1e-6) return false; // collinear outX = (C*E - F*B) / denom; outY = (A*F - D*C) / denom; return true; }
```

7.4 Anchor Placement Rules

The denominator $(A \times E - B \times D)$ equals zero when all three anchors lie on a single straight line (collinear). In this case, the system cannot compute a unique position. To avoid this:

- Place anchors in a triangle — no three anchors should be in a straight line
- Interior angles of the triangle should be between 30° and 150°
- Minimum separation between any two anchors should be at least 1 metre (for indoor demo) or 10 metres (for field deployment)
- An equilateral triangle gives the most uniform position accuracy across the coverage area

8. Software Stack

8.1 ESP32 Firmware (Arduino IDE)

All four ESP32 units are programmed using the Arduino framework. Two separate firmware images are compiled:

- `anchor.ino` — Broadcasts beacon ID every 200ms via ESP-NOW broadcast. Identical for all 3 anchors; `BEACON_ID` constant is changed before each upload (1, 2, or 3).
- `receiver.ino` — Listens for ESP-NOW packets from all 3 known anchor MACs. Reads RSSI, smooths it with moving average, converts to distance, runs trilateration, prints position to Serial.

The Arduino core used is ESP32 Arduino Core v3.x. A noted compatibility change in v3.x is that the `send` callback signature changed from `const uint8_t*` to `const wifi_tx_info_t*`, requiring an updated function signature in the anchor firmware.

8.2 Python Backend (`server.py`)

A Python 3 script running on the base station laptop performs two functions: it reads the Serial output from the mobile tag ESP32, and it serves that data to the web browser over a WebSocket connection.

- Library: `pyserial` for serial communication
- Library: `websockets` for WebSocket server
- The script parses lines matching: `Position → X: [value] m, Y: [value] m`
- On match, it broadcasts JSON: `{"x": 1.87, "y": 2.14}` to all connected browser clients
- WebSocket port: 8765

8.3 Web Dashboard (`index.html`)

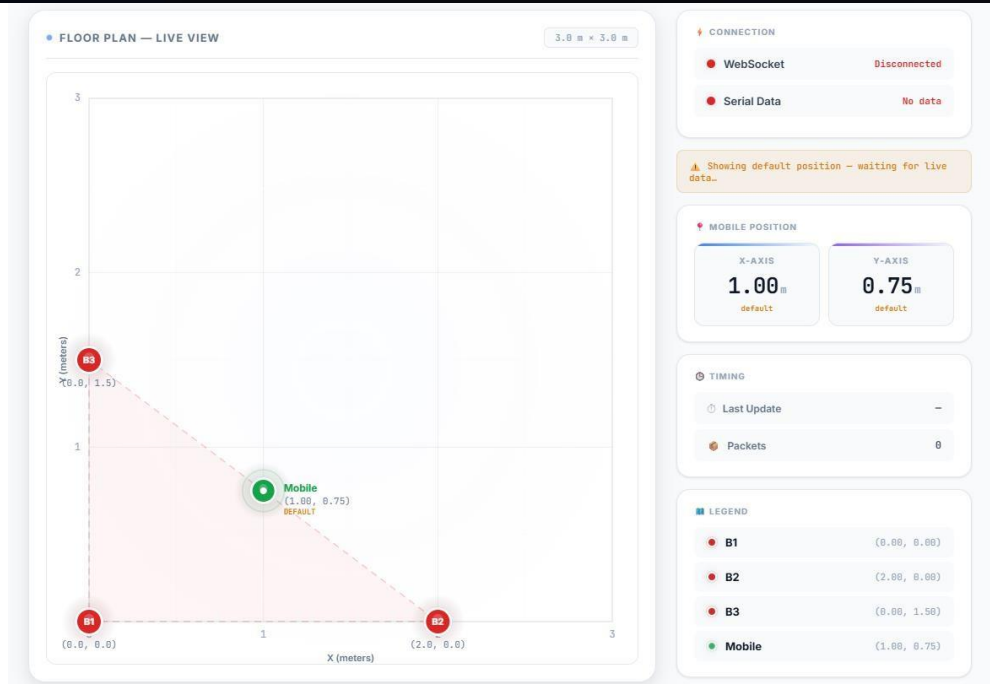
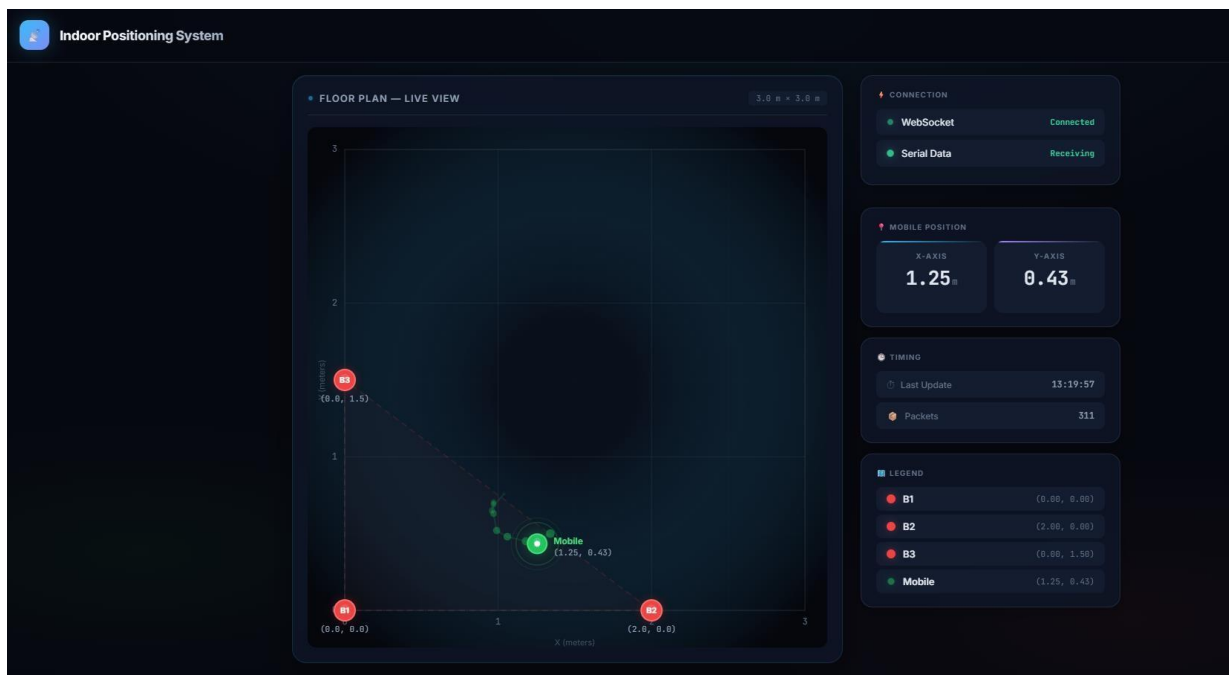
The dashboard is a single self-contained HTML file using vanilla JavaScript and the HTML5 Canvas API. It requires no web server, no npm, and no internet — it is opened directly in a browser. It connects to the Python WebSocket server and renders a live 2D grid map.

9. Real-Time Web Dashboard

9.1 Architecture

The dashboard uses a WebSocket client to receive JSON position updates from the Python server. Rendering is performed on an HTML5 Canvas element at 30 frames per second using `requestAnimationFrame`, decoupled from the WebSocket data ingestion. This ensures smooth animation even when data arrives at irregular intervals.

9.2 Dashboard Elements



PHASE 2: FUTURE WORK

(LoRa + GPS)

12. Why Upgrade to LoRa?

ESP-NOW operates at 2.4 GHz with a range of approximately 200 metres in open conditions. For cattle that may roam several kilometres from the farmer, this is fundamentally inadequate. Phase 2 replaces ESP-NOW with LoRa (Long Range) radio, which is specifically designed for low-power, long-range IoT applications.

13. LoRa Module: SX1278

13.1 Module Overview

The SX1278 is a LoRa transceiver manufactured by Semtech, operating at 433 MHz. It is available as a breakout module (commonly called Ra-02) that connects to the ESP32 via the SPI bus. The module supports transmit power up to +20 dBm and sensitivity down to -148 dBm, giving it outstanding link budget for long-range communication.

13.2 Key Configuration Parameters

- Spreading Factor (SF): 7 — balances range and data rate. Higher SF = longer range but lower throughput.
 - Bandwidth: 125 kHz — standard LoRa bandwidth
 - Coding Rate: 4/5 — error correction overhead
 - Transmit Power: +17 dBm — sufficient for 3–5 km in open fields
-

14. GPS Integration: NEO-8M

14.1 Why GPS on the Anchors (Not the Tags)?

A common question is: why not put GPS directly on each cattle tag and simply read its position? The answer is cost and power. A GPS module draws 25–50 mA continuously and costs ₹600–₹800 per unit. Multiplied across 50 animals, this becomes expensive and impractical from a battery-life standpoint.

Instead, Phase 2 places GPS only on the three anchor nodes — which are carried by farmers who have access to charging — and uses RSSI trilateration for the cattle tags, which can be low-power LoRa transmitters. The GPS provides the anchor coordinates that make trilateration possible in a mobile, field environment.

14.2 NEO-8M Module

The u-blox NEO-8M is a compact, high-sensitivity GPS module with the following characteristics:

- Interface: UART at 9600 baud (connects to ESP32 via Serial2)
- Update rate: 1 Hz (1 position fix per second)
- Cold start time: 26 seconds (time to first fix from power-on)
- Accuracy: 2.5 m CEP (circular error probable) in open sky
- Cost: ₹600–₹800 per module

14.3 GPS Data Processing

The NEO-8M outputs standard NMEA sentences over UART. The ESP32 anchor firmware parses these using the TinyGPS++ library to extract latitude and longitude:

```
#include <TinyGPSPlus.h> TinyGPSPlus gps; if (gps.location.isValid()) { float lat =  
gps.location.lat(); // e.g. 19.076090 float lng = gps.location.lng(); // e.g.  
72.877426 }
```

These GPS coordinates are converted to a local X-Y metric coordinate system using the Haversine formula before being used in the trilateration calculation. Anchor 1 is designated as the local origin (0, 0), and all other positions are expressed in metres relative to it.

14.4 Coordinate Conversion

GPS coordinates are in degrees of latitude and longitude. Trilateration requires distances in metres. The conversion uses:

```
// Convert GPS offset to metres (approximate, valid for small areas) delta_x = (lng2 -  
lng1) × 111320 × cos(lat1 × π/180) delta y = (lat2 - lat1) × 110540
```

This approximation is accurate to within 0.1% for areas up to 10 km × 10 km, which is more than sufficient for cattle grazing ranges.

15. Mobile Anchor Architecture

The most significant architectural difference between Phase 1 and Phase 2 is that anchor positions are no longer fixed and hardcoded — they are measured continuously by GPS and reported dynamically. This section explains exactly how this works.

15.1 The Key Insight

Trilateration requires only two things: the distance from the tag to each anchor, and the position of each anchor. It does not require those anchor positions to be fixed — they simply need to be known at the moment of computation. By equipping each anchor with a GPS module, the anchor can report its own current position alongside the RSSI measurement. The trilateration algorithm then uses the current, real-time positions of the anchors rather than hardcoded constants.

Phase 1 vs Phase 2 — what changes

Phase 1: beacon1X = 0.0, beacon1Y = 0.0 ← hardcoded constants beacon2X = 4.0, beacon2Y = 0.0
beacon3X = 2.0, beacon3Y = 3.0 Phase 2: beacon1X = gps_anchor1.x ← live from GPS, updates every second
beacon2X = gps_anchor2.x beacon3X = gps_anchor3.x Trilateration math is IDENTICAL. Only the coordinate source changes.

15.2 Anchor Node Firmware (Phase 2)

Each Phase 2 anchor node runs a firmware loop that does three things simultaneously:

1. Reads its GPS position every second via TinyGPS++ and stores the latest valid fix
2. Listens for LoRa packets from cattle tags, records the RSSI and the cattle ID
3. Transmits a report packet to the base station containing: anchor ID, anchor GPS coordinates, cattle ID, and RSSI

```
struct AnchorReport {  uint8_t  anchor_id;      // 1, 2, or 3  float  anchor_lat;
// current GPS latitude  float  anchor_lng;      // current GPS longitude  uint16_t
cattle_id;           // which cattle tag was heard  int8_t  rssi;           // signal
strength from that tag };
```

15.3 Base Station Processing (Phase 2)

The base station collects AnchorReport packets from all three anchors. For each cattle tag, it:

4. Converts anchor GPS coordinates to local X-Y metres relative to Anchor 1
5. Applies moving-average smoothing to the RSSI from each anchor for that tag
6. Converts smoothed RSSI to distance using the path loss model
7. Runs the trilateration algorithm using the current anchor positions
8. Broadcasts the result to the dashboard as: {cattle_id, x, y, anchor_positions}

Critically, steps 1–4 run every time new data arrives — meaning if a farmer walks 2 metres to the left, the next trilateration computation will use the updated anchor position automatically.

16. Phase 2 System Data Flow

The complete end-to-end flow of data in Phase 2, from the cattle tag to the farmer's screen, is as follows:

Step 1 — Tag Broadcasts

Each cattle tag (ESP32 + SX1278) wakes from deep sleep every 500 ms, transmits a LoRa packet containing its cattle ID and sequence number, then returns to sleep. This duty cycle keeps current draw extremely low — estimated battery life exceeds 72 hours on a 1000 mAh LiPo.

Step 2 — Anchor Nodes Receive and Report

All three anchor nodes are always in receive mode listening for LoRa packets. When a tag's packet arrives, each anchor records the RSSI and immediately transmits an AnchorReport packet to the base station containing its own current GPS position and the RSSI it measured. Since LoRa range exceeds several kilometres, the base station (carried by one farmer or left at camp) can receive reports from all three anchors.

Step 3 — Base Station Aggregates

The base station Python script collects AnchorReports for each cattle ID. It maintains a state table:

```
cattle_state = { cattle_id: { 'rssi_from_anchor_1': [...buffer...],
'rssi_from_anchor_2': [...buffer...], 'rssi_from_anchor_3': [...buffer...],
'anchor_1_pos': (x, y), # updated from latest GPS 'anchor_2_pos': (x, y),
'anchor_3_pos': (x, y), 'estimated_pos': (x, y) # result of trilateration } }
```

Step 4 — Trilateration

Each time a new AnchorReport arrives, the base station updates the relevant buffers and runs trilateration. If all three anchors have reported within the last 5 seconds, a new position estimate is computed and pushed to the dashboard.

Step 5 — Dashboard Update

The dashboard receives JSON position updates via WebSocket and re-renders the map. In Phase 2, the map background is a satellite tile (from OpenStreetMap or Google Maps static API) with the anchor positions and cattle positions overlaid in real-world GPS coordinates.

Lost Animal Alert Logic

A cattle tag is flagged as 'LOST' if ANY of these conditions are true: 1. No LoRa packet received from that tag in the last 30 seconds 2. RSSI from ALL three anchors drops below -95 dBm simultaneously 3. Estimated position moves outside a configured geofence radius When flagged, the dashboard highlights that animal in red and plays an audio alert.

18. Conclusion

This report has documented the design and implementation of a two-phase cattle monitoring and localization system based on ESP32 microcontrollers and RSSI-based trilateration.

Phase 1, which has been fully implemented and tested, validates the core technical pipeline: three ESP32 anchor nodes broadcast identifiers via ESP-NOW; a mobile ESP32 tag receives these signals, measures RSSI, applies signal smoothing, converts to distance using the log-distance path loss model, and computes its 2D position using trilateration. The computed position is streamed in real time to a web dashboard via a Python WebSocket server. The complete Phase 1 system was built for under ₹1,500 in hardware costs.

Phase 2, planned as the next development stage, replaces ESP-NOW with LoRa (SX1278) radio for kilometre-scale range suitable for real agricultural fields. Anchor nodes are augmented with NEO-8M GPS modules to provide real-time anchor position data, enabling the trilateration algorithm to function with mobile anchors carried by farmers. The trilateration mathematics remain identical to Phase 1 — only the source of anchor coordinates changes from hardcoded constants to live GPS readings.

Together, the two phases describe a complete path from a validated indoor proof of concept to a deployable outdoor cattle monitoring system — at a per-animal hardware cost approximately 70 times lower than existing commercial GPS collar solutions, with no dependency on cellular networks, internet connectivity, or fixed infrastructure.

19. References

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